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An Operating System is the intermediary between a user and the computer's hardware. What the OS does is so important, that without it, a computer would not function properly and would probably collapse. It could be described as an interpreter that translates what the user needs into a language that the hardware understands. However, simply communicating High Level with Low Level is not the only thing an Operating System does. In an ideal world, where there are no complications and only well-intentioned users alive interacting with computers, this might be enough (not guaranteed, but it's a very big possibility). But in the real world, applications become more demanding, code bugs are a thing, memory limitations are real and malicious software as well as malicious people interact with computers too. An Operating System must tackle this by having security, access and task control protocols that handle multiple applications running at once, give access to who needs it (and is allowed to have it), and identifying privileged operations that only it can have access to. Three main Operating Systems (Windows, MacOS and Linux) will be discussed to further explain this. How each of them implements these security and access protocols will also be discussed and compared.

The first OS to be discussed is Windows. Being one of the most popular and used OS (mostly by corporations), it is important to understand how it works and how secure it is. Windows incorporates a strict zero-trust principle to maintain data integrity and safe authentication. Modern OS versions require Unified Extensible Firmware Interface (UEFI), that uses Secure Boot and Trusted Boot protocols that prevent malware from loading onto the system. Secure Boot verifies the digital signature of the framework, while Trusted Boot verifies the digital signature of the kernel. Windows also uses Cryptography (uses keys to make sure that only a specific recipient can read data). It encrypts data using BitLocker following the AES algorithm logic. For authentication, Windows uses multi-factor authentication, biometrics, passcode and in the case of Windows server OS, it implements Kerberos. For access control, Windows implements ACL (access control lists) that basically attaches permissions to files. Thru BitLocker, Secure Boot, Trusted Boot, and thru ACL, Windows ensures confidentiality, integrity and availability for users.

Also gaining popularity (mostly for personal use rather than corporate like Windows), is Apple's macOS. Similar to Windows, macOS implements Secure Boot to protect the system from malware infection at boot time. Setting itself apart from Windows, macOS implements the Secure Enclave protocol, that's basically a secure subsystem that keeps sensitive data protected, even if the kernel would somehow be compromised. Sandboxing is also implemented in the kernel for access control, limiting memory access to users. Apple's macOS also uses what is called the Data Vault that restricts access to the data of an app from other requesting apps. MacOS also uses what they call FileVault, that basically limits other users from decrypting your data or gaining access to your files by only allowing it if your user password is entered. They also use two-factor authentication, touch ID and regular password for user access to the system. Thru FileVault, SecureBoot and Secure Enclave, macOS ensures confidentiality, integrity and availability for users.

Last but not least, another popular OS to be compared to the ones previously mentioned, is Linux. Linux, different from macOS and Windows, is an open-source operating system. Linux uses DAC (resource owner determines who can access their files on the system) and MAC (operating system restricts user access to files based on security labels) access control methods and can also implement Secure Boot, although it can cause malfunctions if the correct drivers are not installed (Secure Boot is mainly Windows). When it comes to authentication, Linux sets itself apart from macOS and Windows. It offers a flexible authentication framework, allowing integration with various authentication services. Compared to the other two, Linux is far more flexible, allowing custom configurations that adjust to user preferences and hence, making it more likable among developers. Thru DAC and MAC implementations, Secure Boot and thru their file system that implements File Hierarchy Standards, Linux ensures confidentiality, integrity and availability for users.

Although all three major Operating Systems are very popular, several differences can be spotted that can set them apart, and, several similarities can glue them together a bit. Although not all Linux implementations use Secure Boot, those that do, align with Windows and macOS principles that also implement Secure Boot as a security measure. All three also implement Advanced Encryption standard (AES) block for data encryption. Windows, adds an additional layer by implementing Trusted Boot as an extra security layer that also checks the OS itself before launching. However, macOS shines as well, by implementing Secure Enclave, that stores data inside, protecting from the operating system, if it would somehow be compromised.

In summary, even though Linux is extremely popular among developers, being open-sourced makes it not be so popular for corporations (Windows mostly succeeds in that department). Apple's macOS rises above in security and reliability by implementing extra layers of memory allocation, data encryption and malware protection, although Windows is not so far behind with Windows 11.

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